Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
plot points on a rectangular coordinate system.			
identify points on a graph.			
verify solutions to an equation in two variables.			
complete a table of solutions to a linear equation.			
find solutions to linear equations in two variables.			

ⓑ *If most of your checks were:*

...confidently. Congratulations! You have achieved the objectives in this section. Reflect on the study skills you used so that you can continue to use them. What did you do to become confident of your ability to do these things? Be specific.

...with some help. This must be addressed quickly because topics you do not master become potholes in your road to success. In math, every topic builds upon previous work. It is important to make sure you have a strong foundation before you move on. Whom can you ask for help? Your fellow classmates and instructor are good resources. Is there a place on campus where math tutors are available? Can your study skills be improved?

...no—I don't get it! This is a warning sign and you must not ignore it. You should get help right away or you will quickly be overwhelmed. See your instructor as soon as you can to discuss your situation. Together you can come up with a plan to get you the help you need.

11.2 Graphing Linear Equations

Learning Objectives

By the end of this section, you will be able to:

- Recognize the relation between the solutions of an equation and its graph
- Graph a linear equation by plotting points
- Graph vertical and horizontal lines

☑ **BE PREPARED** 11.4 Before you get started, take this readiness quiz.

Evaluate: $3x + 2$ when $x = -1$.
If you missed this problem, review Example 3.56.

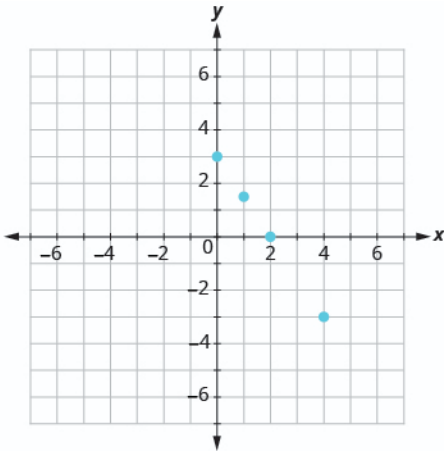
☑ **BE PREPARED** 11.5 Solve the formula: $5x + 2y = 20$ for y.
If you missed this problem, review Example 9.62.

☑ **BE PREPARED** 11.6 Simplify: $\frac{3}{8}(-24)$.
If you missed this problem, review Example 4.28.

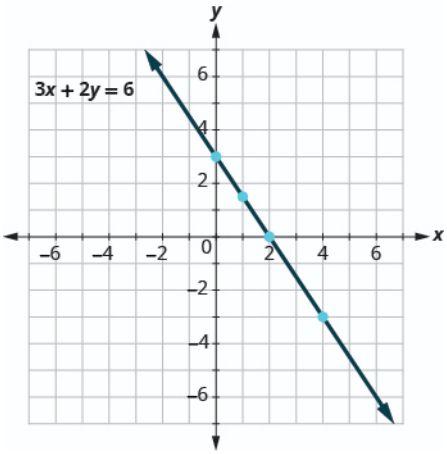
Recognize the Relation Between the Solutions of an Equation and its Graph

In Use the Rectangular Coordinate System, we found a few solutions to the equation $3x + 2y = 6$. They are listed in the table below. So, the ordered pairs $(0, 3)$, $(2, 0)$, $\left(1, \frac{3}{2}\right)$, $(4, -3)$, are some solutions to the equation$3x + 2y = 6$. We can plot these solutions in the rectangular coordinate system as shown on the graph at right.

$3x + 2y = 6$		
x	y	(x, y)
0	3	(0, 3)
2	0	(2, 0)
1	$\frac{3}{2}$	$\left(1, \frac{3}{2}\right)$
4	–3	(4, –3)



Notice how the points line up perfectly? We connect the points with a straight line to get the graph of the equation $3x + 2y = 6$. Notice the arrows on the ends of each side of the line. These arrows indicate the line continues.



Every point on the line is a solution of the equation. Also, every solution of this equation is a point on this line. Points not on the line are *not* solutions!

Notice that the point whose coordinates are $(-2, 6)$ is on the line shown in Figure 11.8. If you substitute $x = -2$ and $y = 6$ into the equation, you find that it is a solution to the equation.

Test (–2, 6):

$$3x + 2y = 6$$
$$3(-2) + 2(6) \stackrel{?}{=} 6$$
$$-6 + 12 \stackrel{?}{=} 6$$
$$6 = 6 \checkmark$$

So (–2, 6) is a solution to the equation.

What about (4, 1)?

$$3x + 2y = 6$$
$$3 \cdot 4 + 2 \cdot 1 \stackrel{?}{=} 6$$
$$12 + 2 \stackrel{?}{=} 6$$
$$14 \neq 6$$

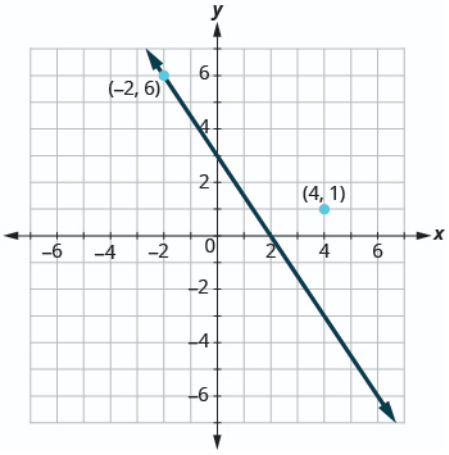


Figure 11.8

So $(4, 1)$ is not a solution to the equation $3x + 2y = 6$. Therefore the point $(4, 1)$ is not on the line.

This is an example of the saying," A picture is worth a thousand words." The line shows you all the solutions to the equation. Every point on the line is a solution of the equation. And, every solution of this equation is on this line. This line is called the *graph* of the equation $3x + 2y = 6$.

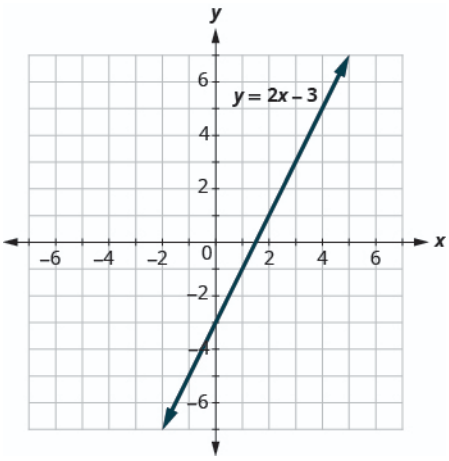
Graph of a Linear Equation

The graph of a linear equation $Ax + By = C$ is a straight line.

- Every point on the line is a solution of the equation.
- Every solution of this equation is a point on this line.

EXAMPLE 11.15

The graph of $y = 2x - 3$ is shown below.



For each ordered pair decide

ⓐ Is the ordered pair a solution to the equation? ⓑ Is the point on the line? (a) $(0, -3)$ (b) $(3, 3)$ (c) $(2, -3)$ (d) $(-1, -5)$

Solution

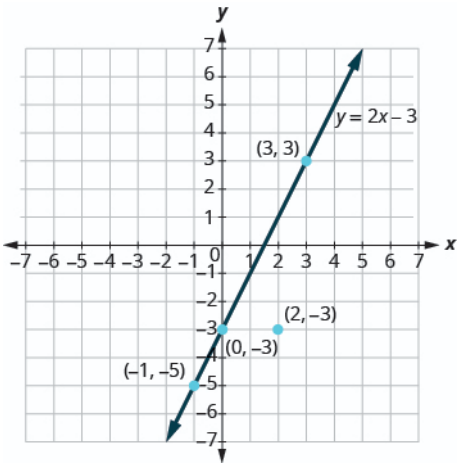
Substitute the x- and y-values into the equation to check if the ordered pair is a solution to the equation.

ⓐ

(a) (0, –3)	(b) (3, 3)	(c) (2, –3)	(d) (–1, –5)
$y = 2x - 3$	$y = 2x - 3$	$y = 2x - 3$	$y = 2x - 3$
$-3 \stackrel{?}{=} 2(0) - 3$	$3 \stackrel{?}{=} 2(3) - 3$	$-3 \stackrel{?}{=} 2(2) - 3$	$-5 \stackrel{?}{=} 2(-1) - 3$
$-3 = -3$ ✓	$3 = 3$ ✓	$-3 \neq 1$	$-5 = -5$ ✓
(0, –3) is a solution.	(3, 3) is a solution.	(2, –3) is not a solution.	(–1, –5) is a solution.

ⓑ Plot the points A: $(0, -3)$ B: $(3, 3)$ C: $(2, -3)$ and D: $(-1, -5)$.

The points $(0, -3)$, $(3, 3)$, and $(-1, -5)$ are on the line $y = 2x - 3$, and the point $(2, -3)$ is not on the line.

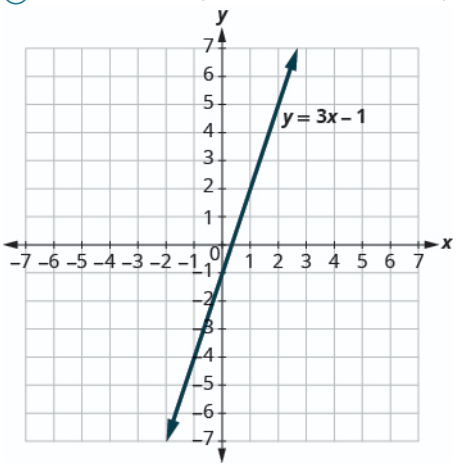


The points which are solutions to $y = 2x - 3$ are on the line, but the point which is not a solution is not on the line.

TRY IT 11.29 The graph of $y = 3x - 1$ is shown.

For each ordered pair, decide

ⓐ is the ordered pair a solution to the equation? ⓑ is the point on the line?



1. $(0, -1)$
2. $(2, 2)$
3. $(3, -1)$
4. $(-1, -4)$

Graph a Linear Equation by Plotting Points

There are several methods that can be used to graph a linear equation. The method we used at the start of this section to graph is called plotting points, or the Point-Plotting Method.

Let's graph the equation $y = 2x + 1$ by plotting points.

We start by finding three points that are solutions to the equation. We can choose any value for x or y, and then solve for the other variable.

Since y is isolated on the left side of the equation, it is easier to choose values for x. We will use 0, 1, and -2 for x for this example. We substitute each value of x into the equation and solve for y.

$x = -2$	$x = 0$	$x = 1$
$y = 2x + 1$	$y = 2x + 1$	$y = 2x + 1$
$y = 2(-2) + 1$	$y = 2(0) + 1$	$y = 2(1) + 1$
$y = -4 + 1$	$y = 0 + 1$	$y = 2 + 1$
$y = -3$	$y = 1$	$y = 3$
$(-2, -3)$	$(0, 1)$	$(1, 3)$

We can organize the solutions in a table. See Table 11.2.

$y = 2x + 1$		
x	y	(x, y)
0	1	$(0, 1)$
1	3	$(1, 3)$
-2	-3	$(-2, -3)$

Table 11.2

Now we plot the points on a rectangular coordinate system. Check that the points line up. If they did not line up, it would mean we made a mistake and should double-check all our work. See Figure 11.9.

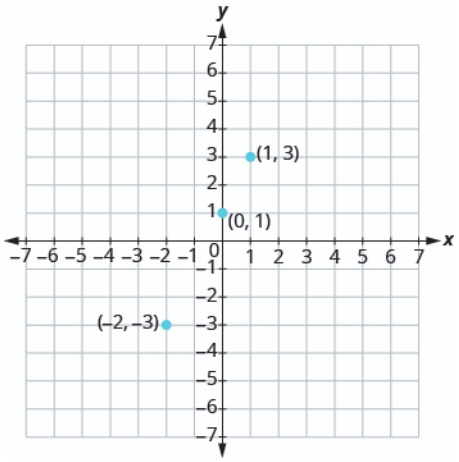


Figure 11.9

Draw the line through the three points. Extend the line to fill the grid and put arrows on both ends of the line. The line is the graph of $y = 2x + 1$.

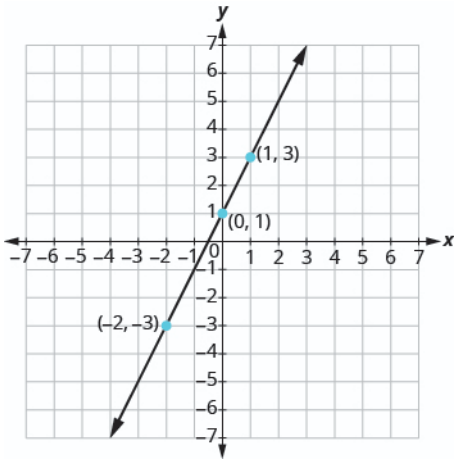


Figure 11.10

HOW TO

Graph a linear equation by plotting points.

Step 1. Find three points whose coordinates are solutions to the equation. Organize them in a table.
Step 2. Plot the points on a rectangular coordinate system. Check that the points line up. If they do not, carefully check your work.
Step 3. Draw the line through the points. Extend the line to fill the grid and put arrows on both ends of the line.

It is true that it only takes two points to determine a line, but it is a good habit to use three points. If you plot only two points and one of them is incorrect, you can still draw a line but it will not represent the solutions to the equation. It will be the wrong line. If you use three points, and one is incorrect, the points will not line up. This tells you something is wrong and you need to check your work. See Figure 11.11.

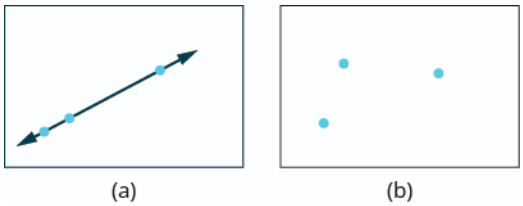


Figure 11.11 Look at the difference between (a) and (b). All three points in (a) line up so we can draw one line through them. The three points in (b) do not line up. We cannot draw a single straight line through all three points.

EXAMPLE 11.16

Graph the equation $y = -3x$.

Solution

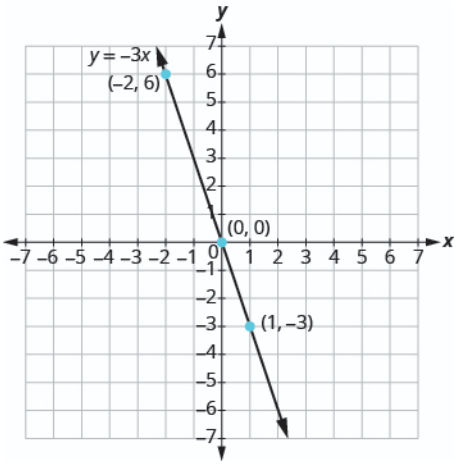
Find three points that are solutions to the equation. It's easier to choose values for x, and solve for y. Do you see why?

$x = 0$	$x = 1$	$x = -2$
$y = -3x$	$y = -3x$	$y = -3x$
$y = -3(0)$	$y = -3(1)$	$y = -3(-2)$
$y = 0$	$y = -3$	$y = 6$
$(0, 0)$	$(1, -3)$	$(-2, 6)$

List the points in a table.

$y = -3x$		
x	y	(x, y)
0	0	$(0, 0)$
1	3	$(1, -3)$
-2	6	$(-2, 6)$

Plot the points, check that they line up, and draw the line as shown.



TRY IT 11.30 Graph the equation by plotting points: $y = -4x$.

TRY IT 11.31 Graph the equation by plotting points: $y = x$.

When an equation includes a fraction as the coefficient of x, we can substitute any numbers for x. But the math is easier if we make 'good' choices for the values of x. This way we will avoid fraction answers, which are hard to graph precisely.

EXAMPLE 11.17

Graph the equation $y = \frac{1}{2}x + 3$.

Solution

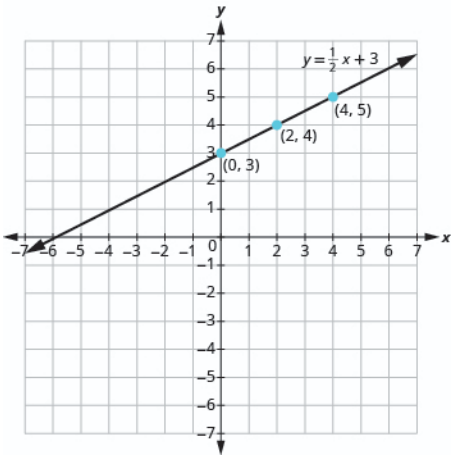
Find three points that are solutions to the equation. Since this equation has the fraction $\frac{1}{2}$ as a coefficient of x, we will choose values of x carefully. We will use zero as one choice and multiples of 2 for the other choices.

$x = 0$	$x = 2$	$x = 4$
$y = \frac{1}{2}x + 3$	$y = \frac{1}{2}x + 3$	$y = \frac{1}{2}x + 3$
$y = \frac{1}{2}(0) + 3$	$y = \frac{1}{2}(2) + 3$	$y = \frac{1}{2}(4) + 3$
$y = 3$	$y = 4$	$y = 5$
$(0, 3)$	$(2, 4)$	$(4, 5)$

The points are shown in the table.

$y = \frac{1}{2}x + 3$		
x	y	(x, y)
0	3	$(0, 3)$
2	4	$(2, 4)$
4	5	$(4, 5)$

Plot the points, check that they line up, and draw the line as shown.



TRY IT 11.32 Graph the equation: $y = \frac{1}{3}x - 1$.

TRY IT 11.33 Graph the equation: $y = \frac{1}{4}x + 2$.

So far, all the equations we graphed had y given in terms of x. Now we'll graph an equation with x and y on the same side.

EXAMPLE 11.18

Graph the equation $x + y = 5$.

Solution

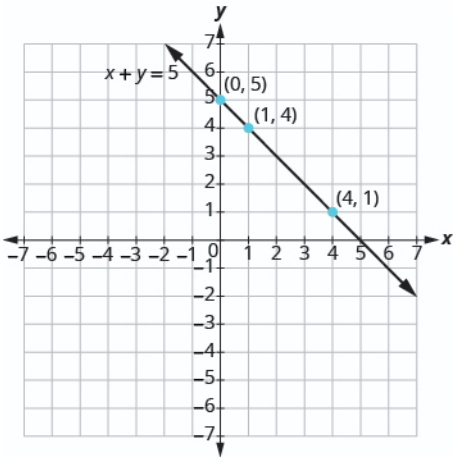
Find three points that are solutions to the equation. Remember, you can start with *any* value of x or y.

$x = 0$	$x = 1$	$x = 4$
$x + y = 5$	$x + y = 5$	$x + y = 5$
$0 + y = 5$	$1 + y = 5$	$4 + y = 5$
$y = 5$	$y = 4$	$y = 1$
$(0, 5)$	$(1, 4)$	$(4, 1)$

We list the points in a table.

$x+y=5$		
x	y	(x,y)
0	5	$(0,5)$
1	4	$(1,4)$
4	1	$(4,1)$

Then plot the points, check that they line up, and draw the line.



TRY IT 11.34 Graph the equation: $x+y=-2$.

TRY IT 11.35 Graph the equation: $x-y=6$.

In the previous example, the three points we found were easy to graph. But this is not always the case. Let's see what happens in the equation $2x+y=3$. If y is 0, what is the value of x?

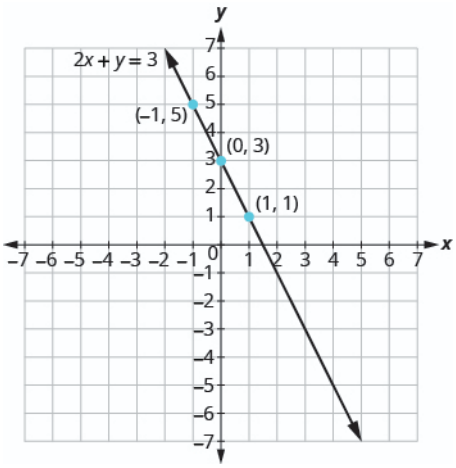
$$2x+y=3$$
$$2x+0=3$$
$$2x=3$$
$$x=\frac{3}{2}$$

The solution is the point $\left(\frac{3}{2},0\right)$. This point has a fraction for the x-coordinate. While we could graph this point, it is hard to be precise graphing fractions. Remember in the example $y=\frac{1}{2}x+3$, we carefully chose values for x so as not to graph fractions at all. If we solve the equation $2x+y=3$ for y, it will be easier to find three solutions to the equation.

$$2x+y=3$$
$$y=-2x+3$$

Now we can choose values for x that will give coordinates that are integers. The solutions for $x=0$, $x=1$, and $x=-1$ are shown.

$y = -2x + 3$		
x	y	(x, y)
0	3	$(0, 3)$
1	1	$(1, 1)$
-1	5	$(-1, 5)$



EXAMPLE 11.19

Graph the equation $3x + y = -1$.

Solution

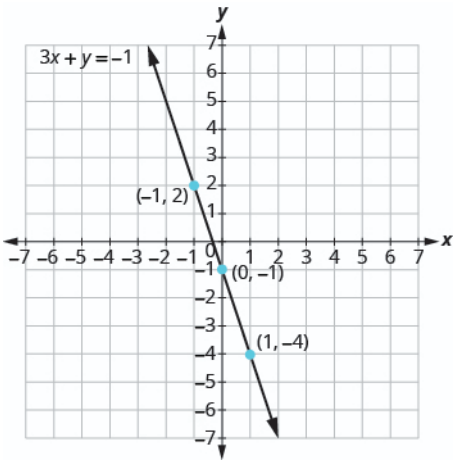
Find three points that are solutions to the equation.

First, solve the equation for y.

$$\begin{aligned} 3x + y &= -1 \\ y &= -3x - 1 \end{aligned}$$

We'll let x be $0, 1$, and -1 to find three points. The ordered pairs are shown in the table. Plot the points, check that they line up, and draw the line.

$y = -3x - 1$		
x	y	(x, y)
0	-1	$(0, -1)$
1	-4	$(1, -4)$
-1	2	$(-1, 2)$



If you can choose any three points to graph a line, how will you know if your graph matches the one shown in the answers in the book? If the points where the graphs cross the x- and y-axes are the same, the graphs match.

TRY IT 11.36 Graph each equation: $2x + y = 2$.

TRY IT 11.37 Graph each equation: $4x + y = -3$.

Graph Vertical and Horizontal Lines

Can we graph an equation with only one variable? Just x and no y, or just y without an x? How will we make a table of values to get the points to plot?

Let's consider the equation $x = -3$. The equation says that x is always equal to -3, so its value does not depend on y. No matter what y is, the value of x is always -3.

To make a table of solutions, we write -3 for all the x values. Then choose any values for y. Since x does not depend on y, you can chose any numbers you like. But to fit the size of our coordinate graph, we'll use $1, 2$, and 3 for the y-coordinates as shown in the table.

$x = -3$		
x	y	(x, y)
-3	1	$(-3, 1)$
-3	2	$(-3, 2)$
-3	3	$(-3, 3)$

Then plot the points and connect them with a straight line. Notice in Figure 11.12 that the graph is a **vertical line**.

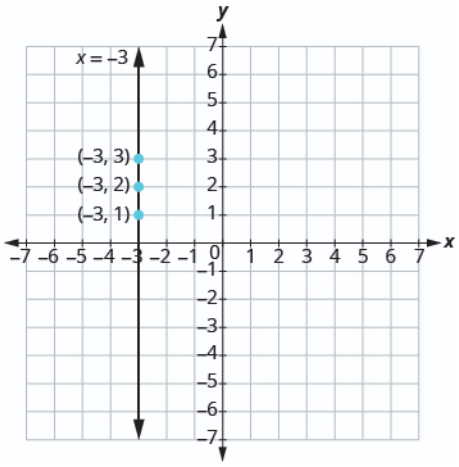


Figure 11.12

Vertical Line

A vertical line is the graph of an equation that can be written in the form $x = a$.

The line passes through the x-axis at $(a, 0)$.

EXAMPLE 11.20

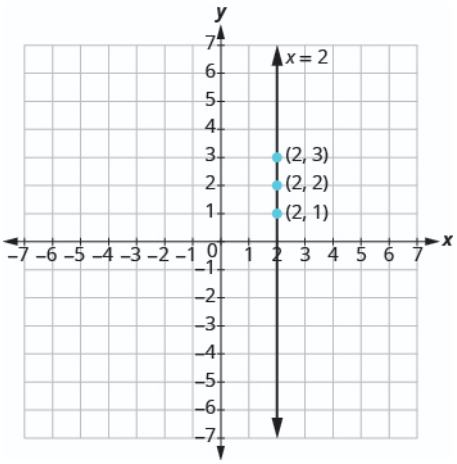
Graph the equation $x = 2$. What type of line does it form?

Solution

The equation has only variable, x, and x is always equal to 2. We make a table where x is always 2 and we put in any values for y.

$x = 2$		
x	y	(x, y)
2	1	$(2, 1)$
2	2	$(2, 2)$
2	3	$(2, 3)$

Plot the points and connect them as shown.



The graph is a vertical line passing through the x-axis at 2.

TRY IT 11.38 Graph the equation: $x = 5$.

TRY IT 11.39 Graph the equation: $x = -2$.

What if the equation has y but no x? Let's graph the equation $y = 4$. This time the y-value is a constant, so in this equation y does not depend on x.

To make a table of solutions, write 4 for all the y values and then choose any values for x.

We'll use $0, 2$, and 4 for the x-values.

$y = 4$		
x	y	(x, y)
0	4	$(0, 4)$
2	4	$(2, 4)$
4	4	$(4, 4)$

Plot the points and connect them, as shown in Figure 11.13. This graph is a **horizontal line** passing through the y-axis at 4.

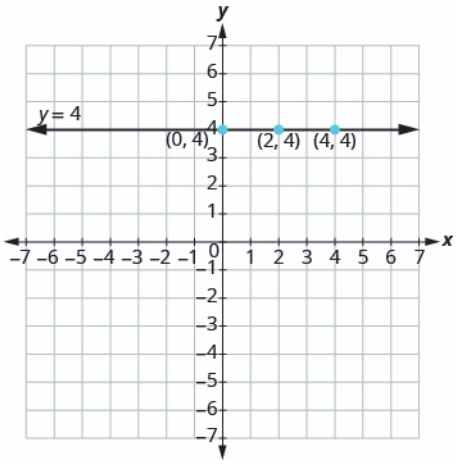


Figure 11.13

Horizontal Line

A horizontal line is the graph of an equation that can be written in the form $y = b$.

The line passes through the y-axis at $(0, b)$.

EXAMPLE 11.21

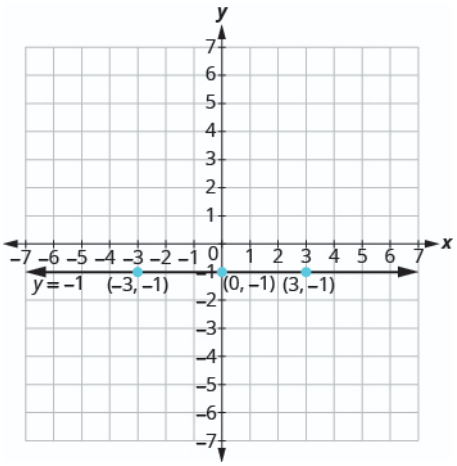
Graph the equation $y = -1$.

Solution

The equation $y = -1$ has only variable, y. The value of y is constant. All the ordered pairs in the table have the same y-coordinate, -1. We choose $0, 3$, and -3 as values for x.

$y = -1$		
x	y	(x, y)
-3	-1	$(-3, -1)$
0	-1	$(0, -1)$
3	-1	$(3, -1)$

The graph is a horizontal line passing through the y-axis at -1 as shown.



TRY IT 11.40 Graph the equation: $y = -4$.

TRY IT 11.41 Graph the equation: $y = 3$.

The equations for vertical and horizontal lines look very similar to equations like $y = 4x$. What is the difference between the equations $y = 4x$ and $y = 4$?

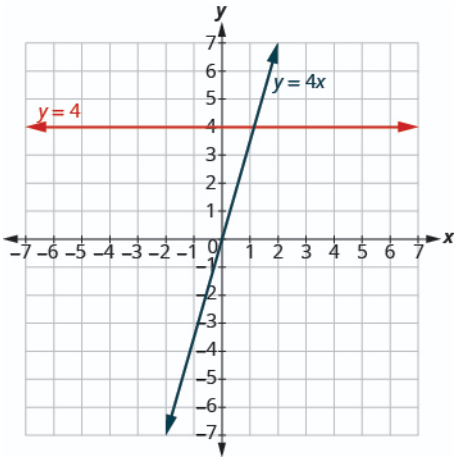
The equation $y = 4x$ has both x and y. The value of y depends on the value of x. The y-coordinate changes according to the value of x.

The equation $y = 4$ has only one variable. The value of y is constant. The y-coordinate is always 4. It does not depend on the value of x.

$y = 4x$		
x	y	(x, y)
0	0	(0, 0)
1	4	(1, 4)
2	8	(2, 8)

$y = 4$		
x	y	(x, y)
0	4	(0, 4)
1	4	(1, 4)
2	4	(2, 4)

The graph shows both equations.



Notice that the equation $y = 4x$ gives a slanted line whereas $y = 4$ gives a horizontal line.

EXAMPLE 11.22

Graph $y = -3x$ and $y = -3$ in the same rectangular coordinate system.

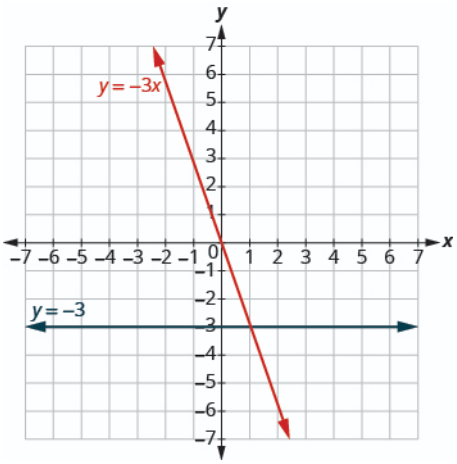
Solution

Find three solutions for each equation. Notice that the first equation has the variable x, while the second does not. Solutions for both equations are listed.

$y = -3x$		
x	y	(x, y)
0	0	(0, 0)
1	–3	(1, –3)
2	–6	(2, –6)

$y = -3$		
x	y	(x, y)
0	–3	(0, –3)
1	–3	(1, –3)
2	–3	(2, –3)

The graph shows both equations.



TRY IT 11.42 Graph the equations in the same rectangular coordinate system: $y = -4x$ and $y = -4$.

TRY IT 11.43 Graph the equations in the same rectangular coordinate system: $y = 3$ and $y = 3x$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Use a Table of Values (https://www.openstax.org/l/24tabofval)

Graph a Linear Equation Involving Fractions (https://www.openstax.org/l/24graphlineq)

Graph Horizontal and Vertical Lines (https://www.openstax.org/l/24graphhorvert)

SECTION 11.2 EXERCISES

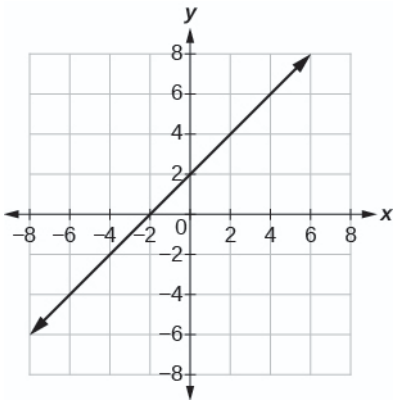
Practice Makes Perfect

Recognize the Relation Between the Solutions of an Equation and its Graph

In each of the following exercises, an equation and its graph is shown. For each ordered pair, decide

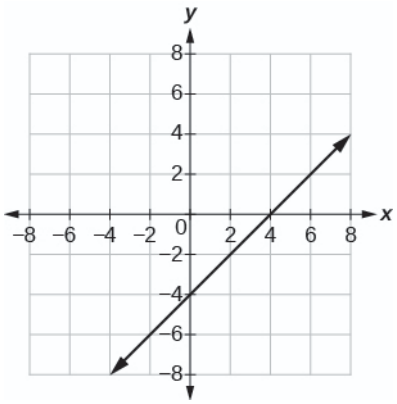
1. ⓐ is the ordered pair a solution to the equation?
2. ⓑ is the point on the line?

39. $y = x + 2$



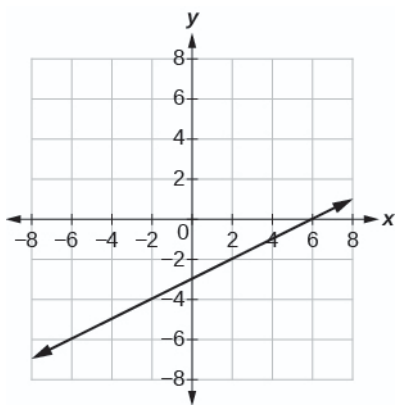
1. $(0, 2)$
2. $(1, 2)$
3. $(-1, 1)$
4. $(-3, 1)$

40. $y = x - 4$



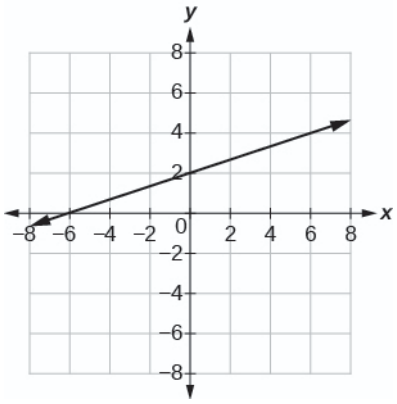
1. $(0, -4)$
2. $(3, -1)$
3. $(2, 2)$
4. $(1, -5)$

41. $y = \frac{1}{2}x - 3$



1. $(0, -3)$
2. $(2, -2)$
3. $(-2, -4)$
4. $(4, 1)$

42. $y = \frac{1}{3}x + 2$



1. $(0, 2)$
2. $(3, 3)$
3. $(-3, 2)$
4. $(-6, 0)$

Graph a Linear Equation by Plotting Points

In the following exercises, graph by plotting points.

43. $y = 3x - 1$

44. $y = 2x + 3$

45. $y = -2x + 2$

46. $y = -3x + 1$

47. $y = x + 2$

48. $y = x - 3$

49. $y = -x - 3$

50. $y = -x - 2$

51. $y = 2x$

52. $y = 3x$

53. $y = -4x$

54. $y = -2x$

55. $y = \frac{1}{2}x + 2$

56. $y = \frac{1}{3}x - 1$

57. $y = \frac{4}{3}x - 5$

58. $y = \frac{3}{2}x - 3$

59. $y = -\frac{2}{5}x + 1$

60. $y = -\frac{4}{5}x - 1$

61. $y = -\frac{3}{2}x + 2$

62. $y = -\frac{5}{3}x + 4$

63. $x + y = 6$

64. $x + y = 4$

65. $x + y = -3$

66. $x + y = -2$

67. $x - y = 2$

68. $x - y = 1$

69. $x - y = -1$

70. $x - y = -3$ **71.** $-x + y = 4$ **72.** $-x + y = 3$

73. $-x - y = 5$ **74.** $-x - y = 1$ **75.** $3x + y = 7$

76. $5x + y = 6$ **77.** $2x + y = -3$ **78.** $4x + y = -5$

79. $2x + 3y = 12$ **80.** $3x - 4y = 12$ **81.** $\frac{1}{3}x + y = 2$

82. $\frac{1}{2}x + y = 3$

Graph Vertical and Horizontal lines

In the following exercises, graph the vertical and horizontal lines.

83. $x = 4$ **84.** $x = 3$ **85.** $x = -2$

86. $x = -5$ **87.** $y = 3$ **88.** $y = 1$

89. $y = -5$ **90.** $y = -2$ **91.** $x = \frac{7}{3}$

92. $x = \frac{5}{4}$

In the following exercises, graph each pair of equations in the same rectangular coordinate system.

93. $y = -\frac{1}{2}x$ and $y = -\frac{1}{2}$ **94.** $y = -\frac{1}{3}x$ and $y = -\frac{1}{3}$ **95.** $y = 2x$ and $y = 2$

96. $y = 5x$ and $y = 5$

Mixed Practice

In the following exercises, graph each equation.

97. $y = 4x$ **98.** $y = 2x$ **99.** $y = -\frac{1}{2}x + 3$

100. $y = \frac{1}{4}x - 2$ **101.** $y = -x$ **102.** $y = x$

103. $x - y = 3$ **104.** $x + y = -5$ **105.** $4x + y = 2$

106. $2x + y = 6$ **107.** $y = -1$ **108.** $y = 5$

109. $2x + 6y = 12$ **110.** $5x + 2y = 10$ **111.** $x = 3$

112. $x = -4$

Everyday Math

113. **Motor home cost** The Robinsons rented a motor home for one week to go on vacation. It cost them $\$594$ plus $\$0.32$ per mile to rent the motor home, so the linear equation $y = 594 + 0.32x$ gives the cost, y, for driving x miles. Calculate the rental cost for driving $400, 800,$ and $1{,}200$ miles, and then graph the line.

114. **Weekly earning** At the art gallery where he works, Salvador gets paid $\$200$ per week plus 15% of the sales he makes, so the equation $y = 200 + 0.15x$ gives the amount y he earns for selling x dollars of artwork. Calculate the amount Salvador earns for selling $\$900$, $\$1{,}600,$ and $\$2{,}000$, and then graph the line.

Writing Exercises

115. Explain how you would choose three x-values to make a table to graph the line $y = \frac{1}{5}x - 2$.

116. What is the difference between the equations of a vertical and a horizontal line?